

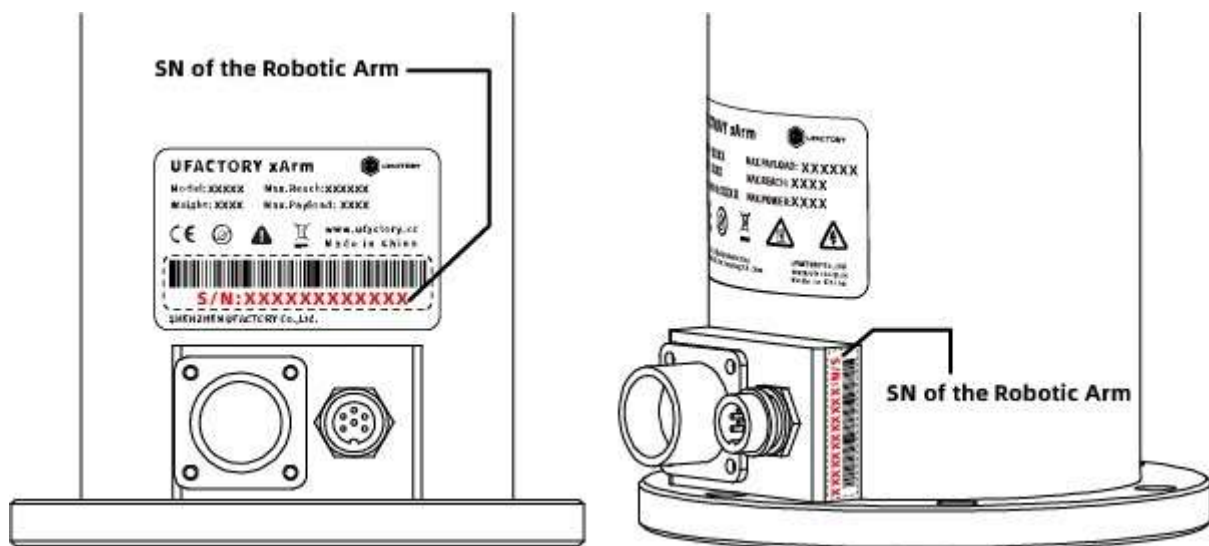
All Collections > xArm-ROS/ Kinematics and Dynamics >
Kinematic and Dynamic Parameters of UFACTORY xArm Series

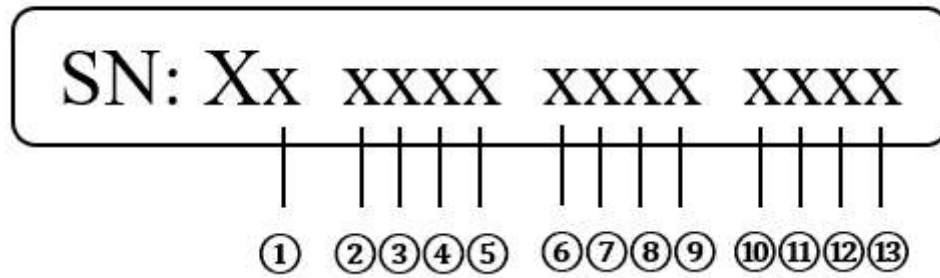
Kinematic and Dynamic Parameters of UFACTORY xArm Series

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Updated over a year ago

The purpose of this article is to guide the user to distinguish the model of the UFACTORY xArm by the SN of the UFACTORY xArm, so that the user can obtain the kinematic and dynamic parameters of the corresponding model of the UFACTORY xArm.

- Method of distinguishing UFACTORY xArm model through SN number





①:

There are three representations for this position: **S/I/F**

S	I	F
xArm 7	xArm 6	xArm 5

⑩:

There are three representations for this position: **L/B/0**

1. When ⑩ position is **0**, Please send the SN of the UFACTORYxArm to technical support (email: support@ufactory.cc).
2. When ⑩ position is **L**, it means that the robotic arm is **model 2**.
3. When ⑩ position is **B**, it means that the robotic arm is **model 1/model 3**

3-1: When the position ⑥⑦ is 04-12 and the position ⑧⑨ is 20, the model of the UFACTORY xArm is **model 3** (such as 0420).

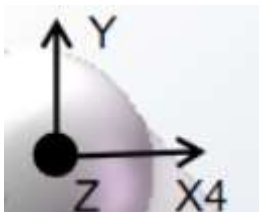
3-2: When the position ⑥⑦⑧⑨ is other numbers, the model of the UFACTORY Arm is **model 1**.

• Kinematic and Dynamic Parameters of UFACTORY xArm

Annotation of coordinate representation:

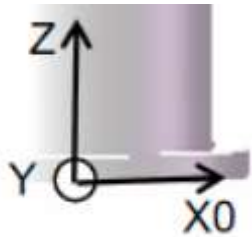
Black dot (●):

This means the direction is pointing out of the viewing surface.



White dot (○):

This means the direction is pointing into the viewing surface.



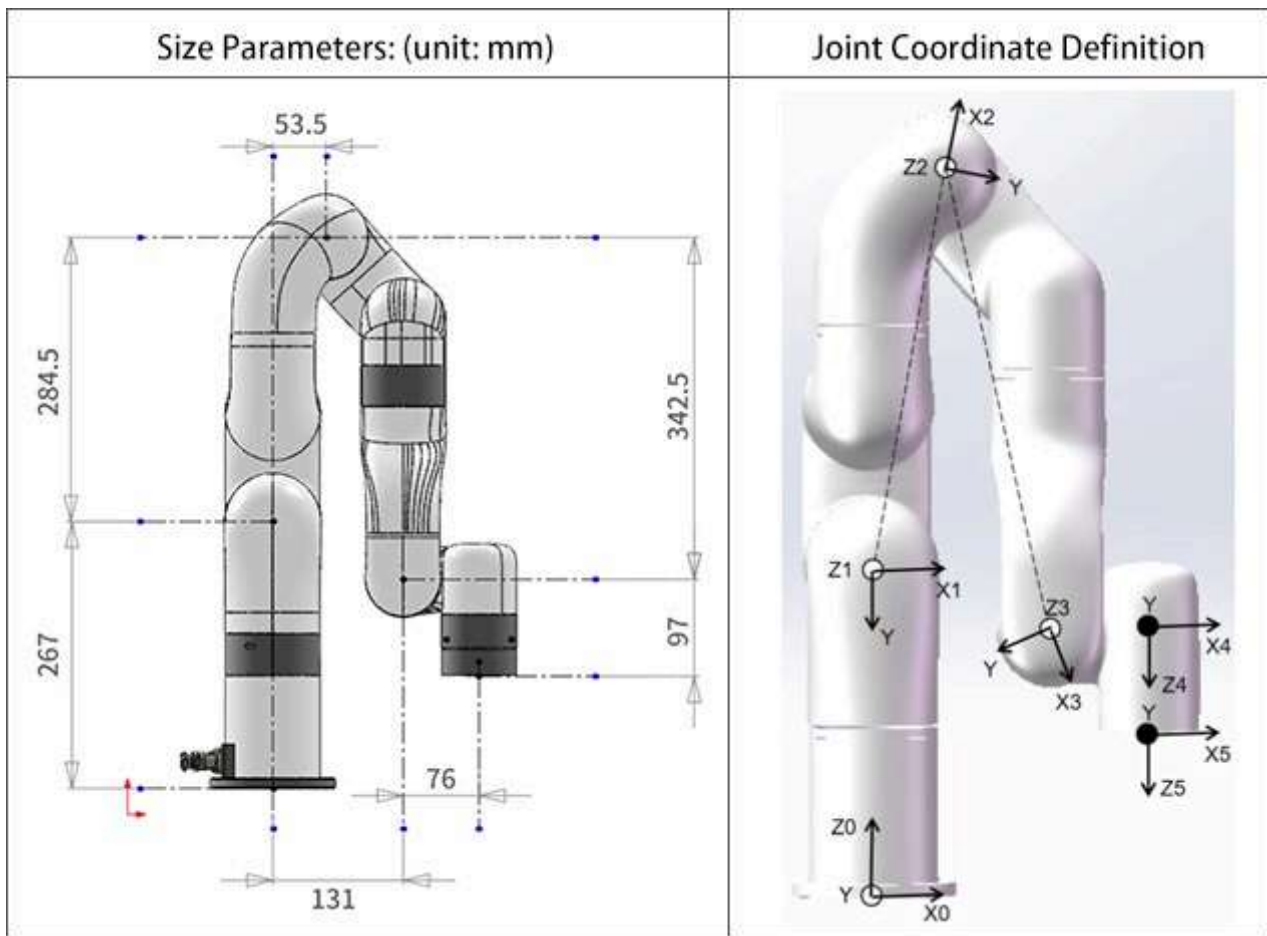
* Please note the provided Dynamic Parameters are just for reference.

1. UFACTORY xArm 5

1) UFACTORY xArm5 Modified D-H Parameters

Size Parameters: (unit: mm)			Joint Coordinate Definition		
Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	0	0	0
Joint2	0	0	$-\pi/2$	0	T2_offset
Joint3	0	0	0	a2	T3_offset
Joint4	0	0	0	a3	T4_offset
Joint5	0	97	$-\pi/2$	76	0

2) UFACTORY xArm5 Standard D-H Parameters



Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	-pi/2	0	0
Joint2	0	0	0	a2	T2_offset
Joint3	0	0	0	a3	T3_offset
Joint4	0	0	-pi/2	76	T4_offset
Joint5	0	97	0	0	0

Note:

'Tx_offset' is the offset joint angle from the mathematical zero position to the mechanical zero position shown in the picture.

$$T2_offset = -\text{atan}(284.5/53.5) = -1.3849179 \text{ } (-79.34995^\circ);$$

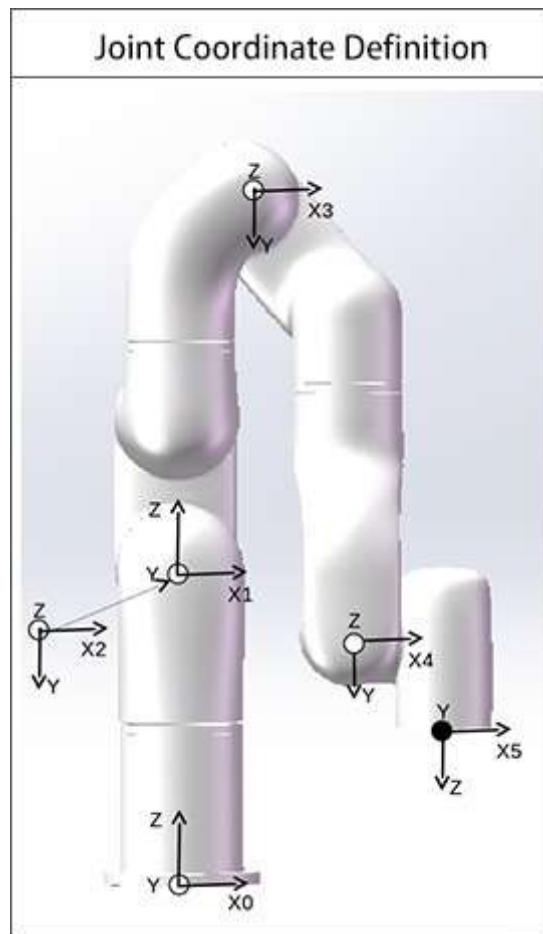
$$T3_offset = \text{atan}(284.5/53.5) + \text{atan}(0.3425/0.0775) = 2.7331843 \text{ } (156.599924^\circ);$$

$$a2 = \text{sqrt}(284.5^2 + 53.5^2) = 289.48866;$$

$$T4_offset = -\text{atan}(342.5/77.5) = -1.3482664 \text{ } (-77.249974^\circ);$$

$$a3 = \text{sqrt}(77.5^2 + 342.5^2) = 351.158796;$$

3) UFACTORY xArm5 xArm Mass Parameters



UFACTORY xArm 5-Model 1

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.177	[0.15, 27.24, -13.57]
Link2	2.011	[36.7, -220.9, 33.56]
Link3	2.01	[68.34, 223.66, 1.1]
Link4	1.206	[63.87, 29.3, 3.5]
Link5	0.17	[0, -6.77, -10.98]

UFACTORY xArm 5-Model 2

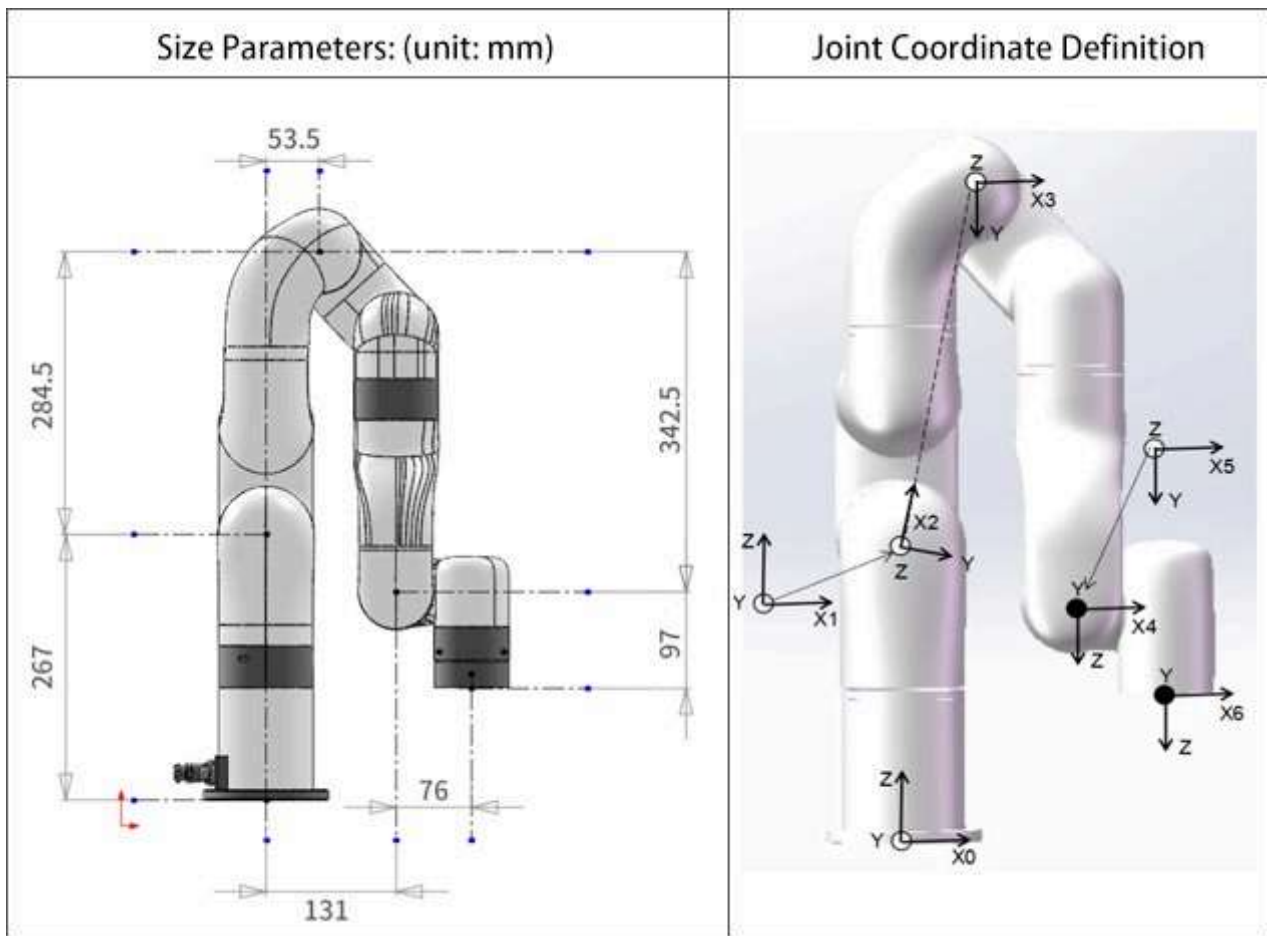
Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.46	[0.13, 30.1, -12.0]
Link2	2.21	[38.2, -226.6, 34.7]
Link3	2.16	[69.0, 231.8, 1.0]
Link4	1.354	[65.2, 31.8, 3.11]
Link5	0.17	[0, -6.77, -10.98]

UFACTORY xArm 5-Model 3

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.382	[0.13, 29.4, -12.4]
Link2	2.164	[37.88, -225.4, 34.47]
Link3	2.121	[68.83, 229.85, 1.02]
Link4	1.317	[64.9, 31.2, 3.2]
Link5	0.17	[0, -6.77, -10.98]

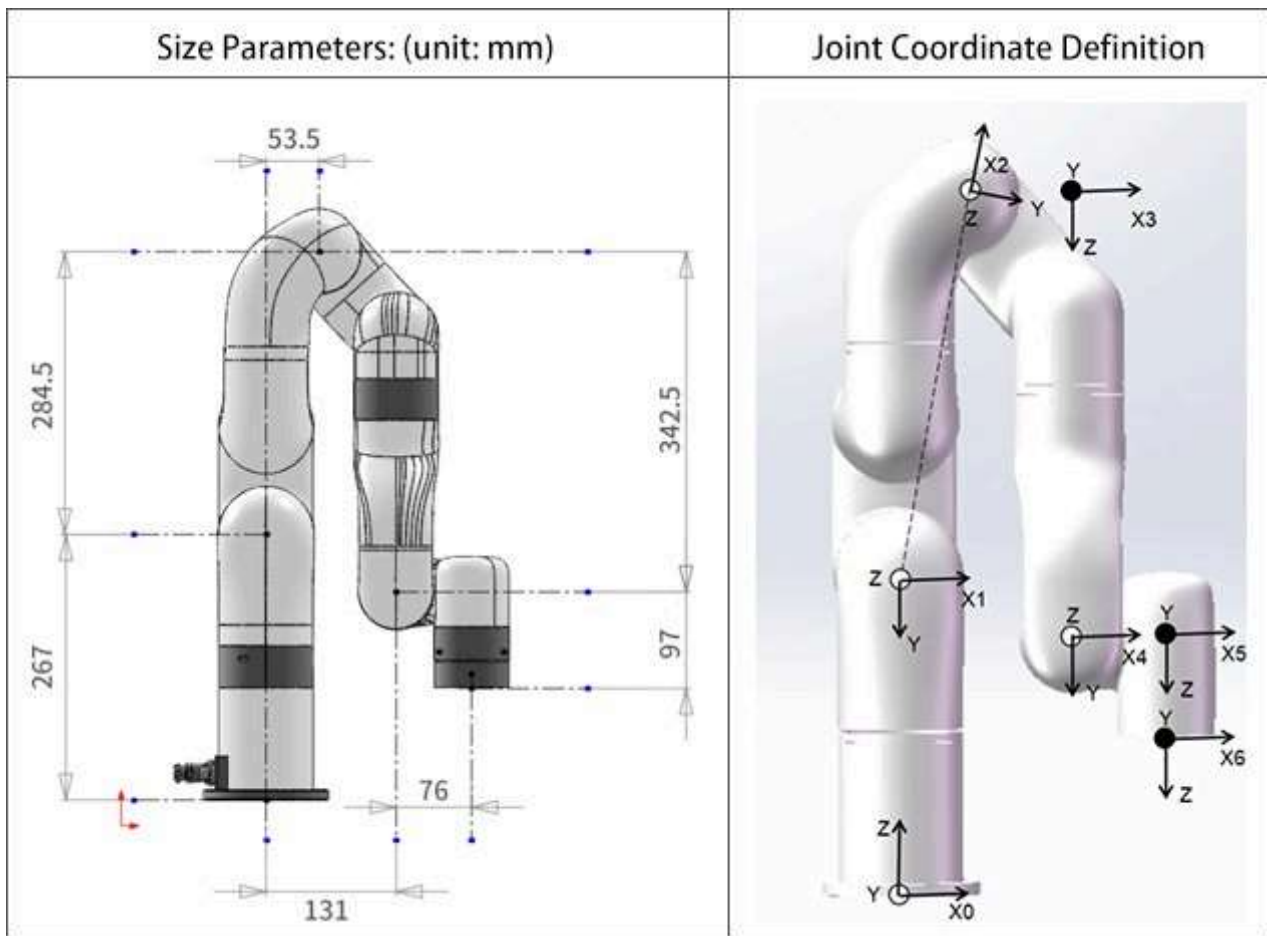
2. UFACTORY xArm 6

1) UFACTORY xArm 6 Modified D-H Parameters



Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	0	0	0
Joint2	0	0	-pi/2	0	T2_offset
Joint3	0	0	0	a2	T3_offset
Joint4	0	342.5	-pi/2	77.5	0
Joint5	0	0	pi/2	0	0
Joint6	0	97	-pi/2	76	0

2) UFACTORY xArm 6 Standard D-H Parameters



Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	-pi/2	0	0
Joint2	0	0	0	a2	T2_offset
Joint3	0	0	-pi/2	77.5	T3_offset
Joint4	0	342.5	pi/2	0	0
Joint5	0	0	-pi/2	76	0
Joint6	0	97	0	0	0

Note:

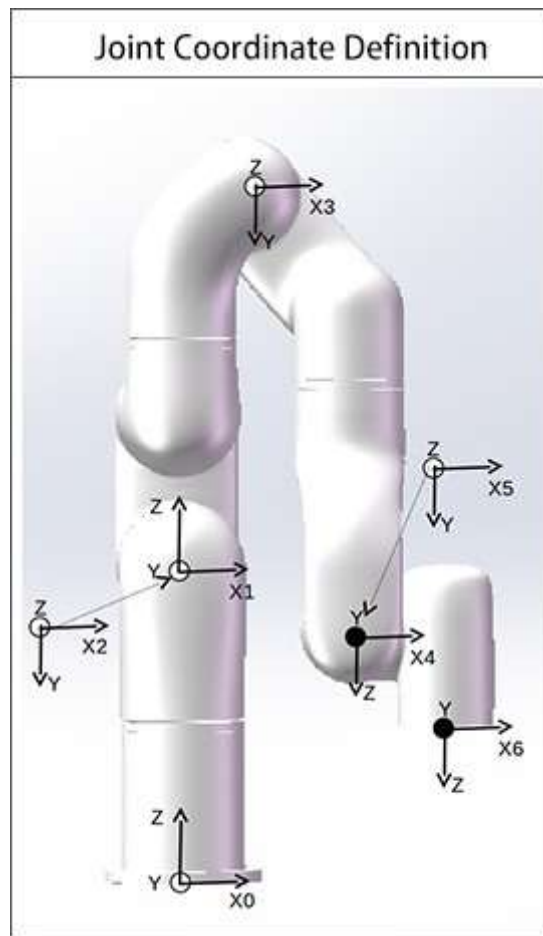
'**Tx_offset**' is the offset joint angle from the mathematical zero position to the mechanical zero position shown in the picture.

$$a2 = \sqrt{284.5^2 + 53.5^2} = 289.48866;$$

$$T2_offset = -\text{atan}(284.5/53.5) = -1.3849179 \text{ } (-79.34995^\circ);$$

$$T3_offset = -T2_offset = 1.3849179 \text{ } (79.34995^\circ);$$

3) UFACTORY xArm 6 Mass Parameters



UFACTORY xArm 6-Model 1

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.177	[0.15, 27.24, -13.57]
Link2	2.011	[36.7, -220.9, 33.56]
Link3	1.725	[69.77, 113.5, 11.6]
Link4	1.211	[-0.2, 20.0, -26.0]
Link5	1.206	[63.87, 29.3, 3.5]
Link6	0.17	[0, -6.77, -10.98]

UFACTORY xArm 6-Model 2

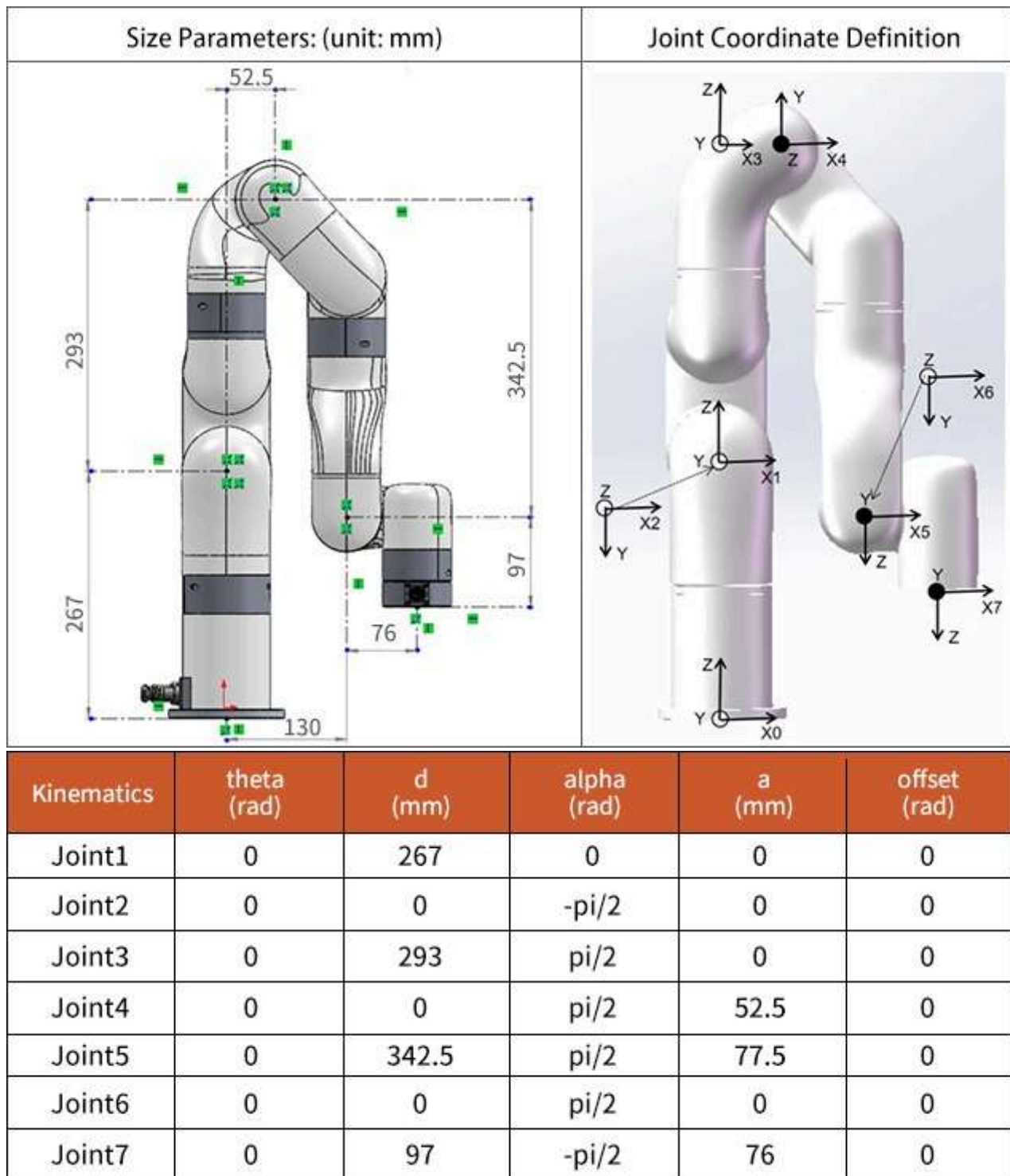
Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.46	[0.13, 30.1, -12.0]
Link2	2.21	[38.2, -226.6, 34.7]
Link3	1.925	[70.6, 117.2, 10.4]
Link4	1.36	[0.18, 17.7, -23.0]
Link5	1.354	[65.2, 31.8, 3.11]
Link6	0.17	[0, -6.77, -10.98]

UFACTORY xArm 6-Model 3

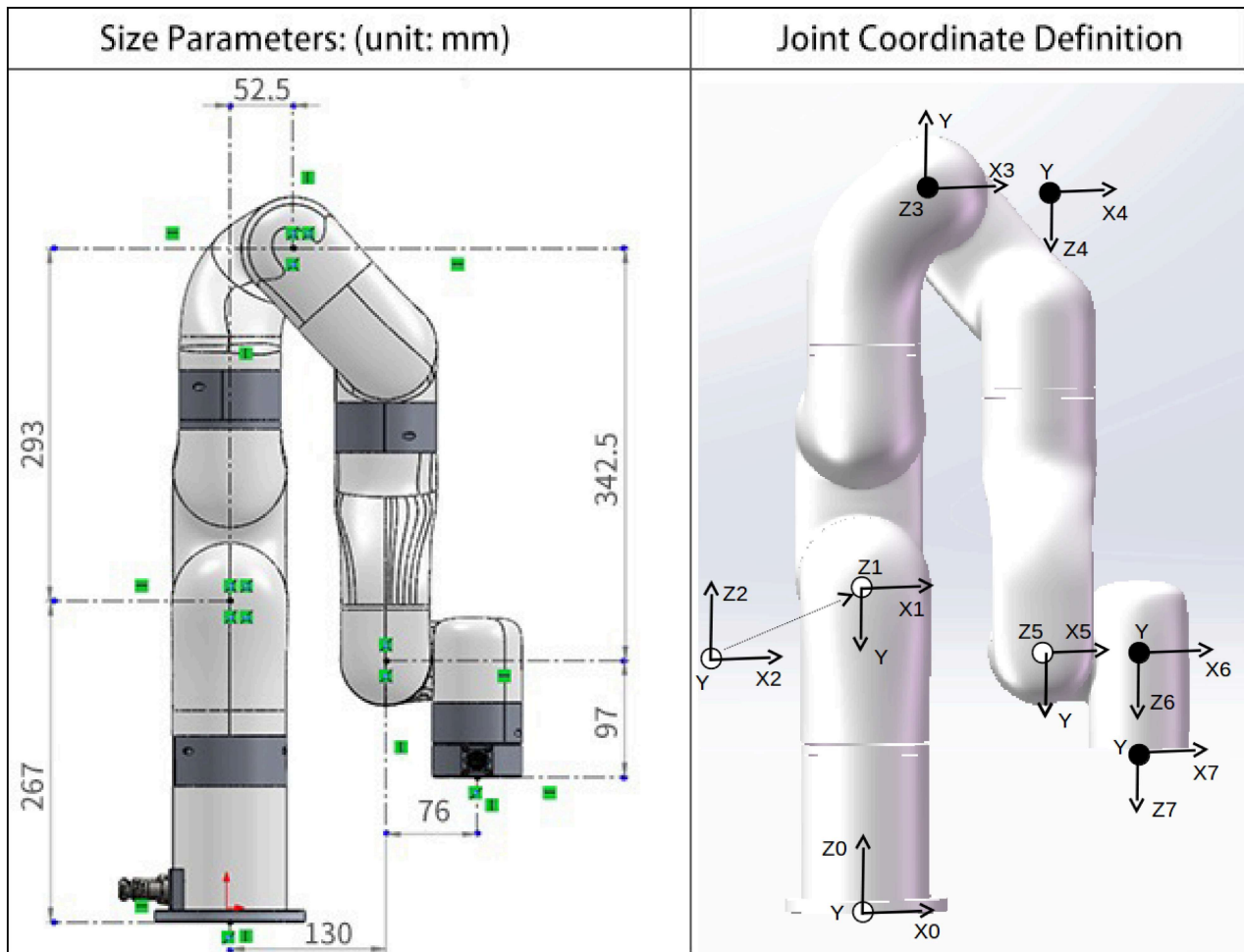
Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.382	[0.13, 29.4, -12.4]
Link2	2.267	[38.8, -227.8, 34.96]
Link3	1.875	[70.4, 116.3, 10.7]
Link4	1.319	[-0.2, 18.0, -22.9]
Link5	1.34	[65.1, 30.96, 3.15]
Link6	0.17	[0, -6.77, -10.98]

3. UFACTORY xArm 7

1) UFACTORY xArm7 Modified D-H Parameters

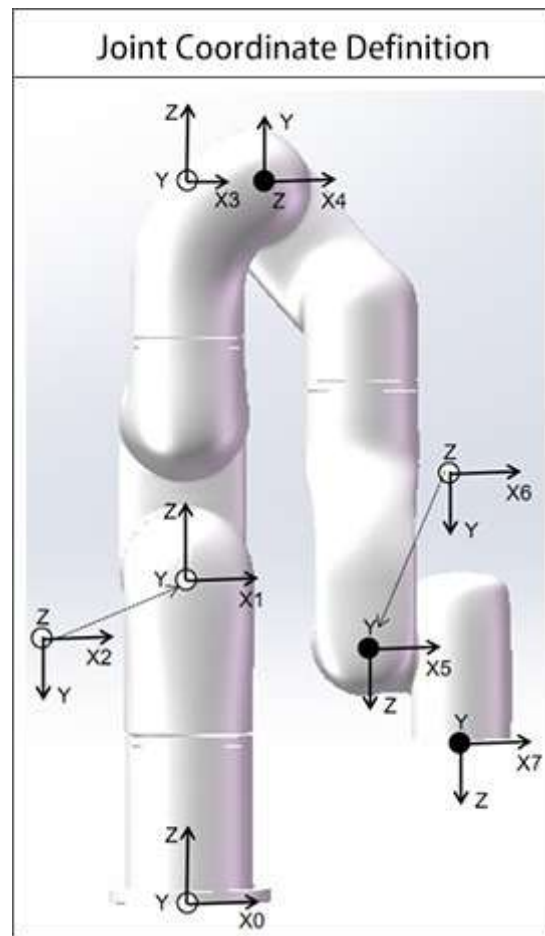


2) UFACTORY xArm7 Standard D-H Parameters



Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	$-\pi/2$	0	0
Joint2	0	0	$\pi/2$	0	0
Joint3	0	293	$\pi/2$	52.5	0
Joint4	0	0	$\pi/2$	77.5	0
Joint5	0	342.5	$\pi/2$	0	0
Joint6	0	0	$-\pi/2$	76	0
Joint7	0	97	0	0	0

3) UFACTORY xArm 7 Mass Parameters



UFACTORY xArm7-Model 1

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.177	[0.15, 27.24, -13.57]
Link2	1.716	[0.22, -124.7, 18.9]
Link3	1.485	[46.0, -22.3, -8.47]
Link4	1.574	[69.75, -112.5, 13.2]
Link5	1.209	[-0.35, 17.6, -28.4]
Link6	1.214	[63.65, 30.84, 21.7]
Link7	0.17	[0, -6.77, -10.98]

UFACTORY xArm7-Model 2

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.46	[0.13, 30.1, -12.0]
Link2	1.916	[0.2, -129.6, 16.9]
Link3	1.69	[46.76, -25.3, -7.46]
Link4	1.774	[70.66, -116.6, 11.7]
Link5	1.357	[-0.3, 15.6, -25.3]
Link6	1.362	[65.0, 33.4, 21.3]
Link7	0.17	[0, -6.77, -10.98]

UFACTORY xArm7-Model 3

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.382	[0.13, 29.4, -12.4]
Link2	1.869	[0.2, -128.56, 17.35]
Link3	1.638	[46.6, -24.63, -7.68]
Link4	1.727	[70.5, -115.75, 12.0]
Link5	1.32	[-0.32, 16.04, -26.0]
Link6	1.325	[64.7, 32.8, 21.4]
Link7	0.17	[0, -6.77, -10.98]

Did this answer your question?



UFACTORY Help Center

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